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Pipe Burst Prevention Relief Valve, Control Valve System and Method

Abstract

A pressure relief valve that is spring-activated or utilizes a burst disk is described for preventing pipe bursts in a building plumbing system. A control valve system and method for preventing pipe bursts utilizing the pressure relief valve are also described. Also described are spring-activated control valve cartridges and control valve systems utilizing spring-activated control valve cartridges that prevent pipe bursts when installed building plumbing systems. Further described are methods for preventing pipe bursts in building plumbing systems comprising the step of disposing a relief valve or spring-activated cartridge according to the disclosure in a building plumbing system.

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Background/Summary

BACKGROUND

[0001] During cold snaps, water in pipes and valves in a building can freeze. When water in pipes and valves freezes, the pipes and valves can burst, damaging the plumbing, electrical and HVAC systems in the building and the area in the vicinity of the burst. Damage associated with pipe bursts can require that the water and other utilities to and systems in the building be turned off, and building occupants evacuate the building and the building remain unoccupied until it has been repaired.

[0002] Repairs associated with a burst pipe or valve can require the replacement of the pipe, valve, and wall in which the pipe or valve was installed, insulation, electrical and HVAC systems, fixtures and finishes in the surrounding building area, and furniture and other possessions in the building. Repairing the building and replacing damaged fixtures, furniture and possessions can cost a considerable amount of money. In addition, the inability to use the pipe or valve, and surrounding area can cause downtime, inconvenience and possibly additional costs associated with loss of use of the building, including rental income, and moving into and renting a temporary space until the repairs are completed.

[0003] It has been reported that burst pipes due to freezing cause over USD \$100 million in the United States and approximately USD \$1 billion worldwide in damages each year. The Red Cross has reported that more than 250,000 families across the United States experience frozen or burst pipes each year. In addition, losses due to burst pipes and valves are on the rise because of increases in the height of apartment buildings, the number of bathrooms per bedroom in dwellings, the number of water-using appliances in buildings, and the amount of water used per person. Losses due to burst pipes and valves are also rising because of the vulnerability of buildings to damage due to modern methods of building construction, such as tiny homes, modular buildings, bathrooms and kitchens, and mass timber or exposed beam construction.

[0004] Bursting pipes and valves in a building due to the freezing of water in them is a problem that persists to this day, despite efforts by others to solve the problem. What is needed, therefore, is a relief valve that can be installed in an existing building plumbing system which will protect it from bursting when pipes or valves in the system freeze. What is also needed is a compact water delivery control valve system for use as a plumbing fixture in a building plumbing system that does not burst and protects the pipes and other fixtures connected to it from bursting under freezing conditions. What is further needed is a method for preventing pipes or valves in a building plumbing system from freezing by disposing the relief valve or valve system having the relief valve in the building plumbing system.

SUMMARY

[0005] Accordingly, in one aspect, a relief valve for pipes or valves or other fixtures in a building plumbing system is described that includes a body having an inlet, an outlet, an exterior surface, and an interior surface that defines a generally longitudinal passageway that extends from the inlet to the outlet, a valve seat disposed on the interior surface of said body between the inlet and the outlet, a flow limiter disposed on the interior surface of said body between said valve seat and the outlet, and extending into the passageway, and a valve seat sealing portion disposed in fluid communication with said valve seat between the inlet and said flow limiter, and configured to seal said valve seat against the flow of fluid in the passageway when the fluid has a pressure that is less than a predetermined pressure. In another aspect, a compact, pipe-burst-prevention valve system for use in a building plumbing system is described that includes a faucet housing having a handle and an outlet for delivering fluid to an end user, a control valve having an inlet and an outlet in fluid communication with each other and the control valve outlet in fluid communication with the faucet outlet, and the relief valve. Further described is a method for preventing pipes from bursting that includes disposing the pressure relief valve or valve system having the pressure relief valve in fluid communication with a building plumbing system.

[0006] Accordingly, in another aspect, a pipe-burst-prevention fluid control valve cartridge for fluid having a pressure is described that includes a body having a stem end, a valve end capable of being held in fluid communication with a fluid control valve of a faucet in a plumbing system, an exterior surface, and a generally cylindrical interior surface, a generally cylindrical split stem, at least a portion which is configured to be rotatably held within the interior of said cartridge body, having a spindle configured to receive a faucet handle, an upper stem sized so as to be received within the interior surface of said cartridge body and configured to be capable of being rotated when the spindle is turned, and an upper end adjacent to the spindle and a lower end generally opposite the upper end and adjacent to the cartridge body valve end with a cavity extending therefrom into the upper stem toward the upper stem upper end, the upper stem configured to extend toward the valve end of the cartridge body when the spindle is rotated to close the valve and retract from the valve end when the spindle is rotated to open the valve, a lower stem having an upper end configured so that a portion of it can be slidably received into the interior of said cartridge body at its valve end and adjacent to the lower end of the upper stem with a piston affixed thereto, and a spring held in the cavity by the piston that is capable of compressing when the pressure of the fluid in the valve equals or exceeds a predetermined pressure, wherein the piston, cavity and spring have diameters such that the cavity can receive slidably the spring and the piston and lengths such that the lower end of the upper stem and the upper end of the lower stem form a gap when the spring is fully extended. In another aspect, a compact, pipe-burst-prevention control valve system for use with fluid having a pressure in a building plumbing system is described that includes a faucet housing having a handle and an outlet for delivering fluid from the building plumbing system, a control valve having an inlet and an outlet in fluid communication with each other and the control valve outlet in fluid communication with the faucet housing outlet, and the split stem cartridge disposed in fluid communication with said control valve within said faucet housing in between the faucet inlet and the faucet outlet. Further described is a method of preventing pipes in a building plumbing system from bursting, comprising the step of disposing a fluid control valve system for use with a fluid having a pressure in fluid communication with the building plumbing system that includes a faucet housing having a handle and an outlet for delivering fluid from the building plumbing system, a control valve having an inlet and an outlet in fluid communication with each other and the control valve outlet in fluid communication with the faucet housing outlet, and the split stem cartridge disposed in fluid communication with said control valve within said faucet housing in between the faucet inlet and the faucet outlet.

[0007] In yet another aspect, a pipe-burst-prevention fluid control valve cartridge for fluid having a pressure is described that includes a body having a stem end, a valve end capable of being held in fluid communication with a fluid control valve of a faucet in a plumbing system, an exterior surface, and a generally cylindrical interior surface, a generally cylindrical split stem, at least a portion of which is configured to be rotatably held within the interior of said body, having a spindle configured to receive a faucet handle that includes an upper stem sized so as to be received within the interior surface of said body and configured to be capable of being rotated when the spindle is turned, and having an upper end adjacent to the spindle and a lower end generally opposite the upper end and adjacent to the cartridge body valve end with the upper end toward the cartridge body valve end with an interior cavity extending therefrom into the upper stem toward the upper stem upper end, a lower stem having an upper end adjacent to the lower end of the upper stem with a piston affixed on the lower stem upper end and a lower end generally opposite the lower stem upper end, the lower stem configured to be slidably received into and rotatably held in the interior of said cartridge body to allow it to rotate when the spindle is rotated and prevent the flow of fluid from the inlet to the outlet when the spindle is rotated to its closed position and the pressure of the fluid is less than the predetermined pressure, and allow the flow of fluid through the outlet when the spindle is rotated to its open position or the spindle is in its closed position and the pressure of the fluid equals or exceeds the predetermined pressure, and a spring held in the cavity by the piston

that is capable of compressing when the pressure of the fluid in the valve equals or exceeds the pre-determined pressure, wherein the piston, cavity and spring have diameters such that the cavity can receive slidably the spring and the piston, and lengths such that the lower end of the upper stem and the upper end of the lower stem form a gap when the spring is fully extended, and a cartridge base held in said cartridge body at its valve end having an opening adjacent to the lower end of the lower stem. In another aspect, a compact, pipe-burst-prevention control valve system for use with fluid having a pressure in a building plumbing system is described that includes a faucet housing having a handle and an outlet for delivering fluid from the building plumbing system, a control valve having an inlet and an outlet in fluid communication with each other and the control valve outlet in fluid communication with the faucet housing outlet, and the cartridge disposed in fluid communication with said control valve within said faucet housing in between the faucet inlet and the faucet outlet. Further described is a method of preventing pipes in a building plumbing system from bursting, comprising the step of disposing the fluid control valve system having the split stem cartridge for use with a fluid having a pressure in fluid communication with the building plumbing system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- [0008] FIG. 1 is a cross-sectional view taken along the longitudinal centerline of one embodiment of a pressure relief valve according to the disclosure;
- [0009] FIG. 2 is a cross-sectional view taken along the longitudinal centerline of an alternative embodiment of a pressure relief valve according to the disclosure;
- [0010] FIG. 3 is a cross-sectional view taken along the longitudinal centerline of a further embodiment of a pressure relief valve according to the disclosure;
- [0011] FIG. 4 is a perspective view of one embodiment of a valve system having a relief valve shown in FIG. 1, 2 or 3, according to the disclosure;
- [0012] FIG. 5 is a flow diagram depicting the flow of fluid through a valve system according to FIG. 4;
- [0013] FIG. 6 is a perspective view of an alternative embodiment of a valve system according to the disclosure;
- [0014] FIG. 7 is a flow diagram depicting the flow of fluid through a valve system according to FIG. 6;
- [0015] FIG. 8 is an exploded perspective view of another embodiment of a valve system according to the disclosure;
- [0016] FIG. 9 is a cross-sectional view taken along the longitudinal centerline of one embodiment of a control valve cartridge according to the disclosure;
- [0017] FIGS. 10A, 10B and 10C depict cross-sectional views of the cartridge embodiment shown in FIG. 9 with the split stem in different positions;
- [0018] FIG. 11 is a cross-sectional view taken long the longitudinal centerline of another embodiment of a control valve cartridge according to the disclosure;
- [0019] FIGS. 12A and 12B depict cross-sectional views of the cartridge embodiment shown in FIG. 11 with the split stem in different positions;
- [0020] FIG. 13 is a perspective view of the lower stem of the cartridge embodiment shown in FIGS. 11, 12A and 12B; and
- [0021] FIG. 14 is an exploded perspective view of another embodiment of a valve system according to the disclosure.

DETAILED DESCRIPTION

Relief Valve

[0022] A pressure relief valve is described that is configured so that it can be installed in an existing building plumbing system to allow fluid to flow through the relief valve to protect the pipes, one or more valve control systems used in faucets, or other fixtures installed in the plumbing system from bursting when the fluid in the pipes, valve systems or other fixtures in the system freezes. The relief valve has a generally tubular and longitudinal body having an interior defining a generally longitudinal passageway through which fluid may flow. The body also has an inlet into which fluid from a high-pressure source is received, an outlet through which fluid at a lower pressure exits the relief valve, and an exterior surface that is generally longitudinal. The relief valve also has a valve seat disposed on the interior surface of the body between the inlet and the outlet, a flow limiter disposed on the interior surface of the body between the valve seat and the outlet and extends into the passageway, and a valve seat sealing portion disposed in fluid communication with the valve seat between the inlet and the flow limiter. The valve seat sealing portion is configured to seal against the valve seat so that the flow of fluid through the valve seat and, accordingly, the passageway, is prevented when the fluid has a pressure that is equal to or less than a predetermined pressure, and not seal against the valve seat, thereby allowing fluid to flow through the valve seat and passageway, when the fluid has a pressure that is great than the predetermined pressure.

[0023] The valve seat sealing portion can be generally transverse or parallel to the exterior surface of the body. In the embodiment in which the valve sealing portion is generally transverse to the exterior surface of the relief valve body, the valve seat is generally parallel to the exterior surface of the body. In the embodiment in which the valve sealing portion is generally parallel to the exterior surface of relief valve body, the valve seat is generally transverse to body exterior.

[0024] An embodiment of the pressure relief valve according to the disclosure that has a valve seat sealing portion that is generally transverse to the exterior surface of the body is the spring-activated pressure relief valve embodiment **100** having a valve seat sealing portion **102** that has a spring **104**, and a generally tubular longitudinal body **106**, shown in FIG. **1**. In this embodiment, body **106** has an exterior surface **108** to which valve seat sealing portion **102** is attached generally transverse to body exterior surface **108**.

[0025] Valve seat sealing portion **102** may be attached using conventional means such as soldering or using screws **110**. The valve seat sealing portion **102** also is preferably attached to exterior surface **108** using a seal or gasket **112** to prevent the flow of fluid between valve seat sealing portion **102** and exterior surface **108**.

[0026] Body **106** has an interior surface **114** that defines a generally longitudinal passageway **116** extending through the interior surface **114** of body **106** through which fluid may flow. The body **106** also has an inlet **118** into which fluid from a high-pressure source in a building plumbing system is received, and an outlet **120** through which fluid flows through passageway **116** out of relief valve **100**.

[0027] Relief valve **100** also has a valve seat **122** disposed on interior surface **114** of body **106** between inlet **118** and outlet **120** and configured to be capable of allowing fluid to flow from inlet **118** to outlet **120** through passageway **116**. Valve seat **122** can be any shape but preferably, for ease of manufacturing, it has a generally cylindrically shaped opening through which fluid may flow. It preferably is constructed integrally with valve body **106**. Alternatively, it may be comprised of a layer of a material that is different from the body **106** applied to interior surface **114** and glued or molded in place.

[0028] Relief valve **100** further has a flow limiter **124** disposed on interior surface **114** between valve seat **122** and outlet **120** and extending into passageway **116**. The flow limiter **124** is sized such that the flow of fluid from inlet **118** to outlet **120** is reduced from a normal flow of fluid in a building plumbing system which is from about 0.5 to about 2.5 gallons per minute (“GPM”) at a normal fluid pressure which for a domestic building plumbing system for water is typically about 40 pounds per square inch (“PSI”). Preferably, flow limiter **124** is sized to reduce the flow of fluid from inlet **118** to outlet **120** by about 50% to about 80% of the normal flow, or from about 0.25 to

about 1.25 GPM and from about 0.10 to about 0.50 GPM. More preferably, flow limiter **124** is sized to limit the flow of fluid through relief valve **100** to less than about 0.4 GPM.

[0029] Flow limiter **124** may be comprised of a single circular opening. In this case, it can have a diameter of less than about 0.05 inches or an opening area of about 0.0020 square inches or less. Preferably, flow limiter **124** has a diameter of between about 0.015 inches to about 0.05 inches or an area of between about 1.77×10^{-4} to about 0.0020 square inches.

[0030] Preferably, flow limiter **124** may be comprised of multiple smaller, circular orifices. A flow limiter **124** comprised of multiple smaller, circular orifices reduces the chance that the flow limiter will become clogged. Where flow limiter **124** is comprised of multiple orifices, the sum of area of the orifices equals approximately that for a flow limiter having a single circular opening.

[0031] Valve seat sealing portion **102** has a movable shaft **126** on which a valve seat holder **128** which is disposed at one end closest to and capable of being held against valve seat **122**. Valve seat sealing portion **102** has a longitudinal center **130** which extends through it and spring holders **132** and **134** each with centers **136** and **138**, respectively, along centerline **130**. Shaft **126** is disposed through spring holder center **138**, through spring **104**, and spring holder center **136**. At least one of spring holders **132** or **134** may have threads **140**, shown disposed on the inside center of holder **132**, and shaft **126** may have threads **142** such that shaft **126** may be rotated to travel inward or outward, thereby adjusting the force of spring **104** to apply a greater or lesser force on shaft **126** than the force of spring **104** alone would apply to shaft **126** without adjustment, and transferring that force to valve seat holder **128** and valve seat **122** to close relief valve **100** and prevent fluid from flowing through it. FIG. 1 shows the interior center **136** of spring holder **132** and the exterior of shaft **126** as having threads on each **140** and **142**, respectively.

[0032] Valve seat sealing portion **102** has a bonnet **144** which is attached sealably to exterior surface **108** of relief valve **100** over centerline **130**. Bonnet **144** has a center **146** which aligns with centerline **130**, and an interior surface **148** opposite valve seat holder **128**.

[0033] As shown in FIG. 1, bonnet **144** holds spring holder **134** against bonnet interior surface **148** on one end opposite valve seat **122** under the force of spring **104** held at the other end adjacent to valve seat **122** by spring holder **132**. Valve seat sealing portion **102** has a collar **150** with a center **152** upon which spring holder **132** rests against the force of spring **104**. Where valve seat sealing portion **102** has two spring holders, as shown in FIG. 1, bonnet **144** holds spring holder **134** against bonnet interior surface **148** under the force of spring **104** with spring holder **132** holding spring **104** at its other end against collar **150**. Valve seat sealing portion **102** need not have spring holder **134**, but rather spring **104** may rest against bonnet interior **148**. Alternatively, valve seat sealing portion **102** need not have spring holder **132**, in which case spring **104** may rest on collar **150** at one end and spring holder **134** at the other end, and spring holder **134** may have threads on its interior and movable shaft **126** may have threads disposed on its exterior adjacent to the threads on the interior of spring holder **134** thereby allowing the force applied to valve seat holder **128** against valve seat **122** to be adjusted.

[0034] Valve seat holder **128** has a flange portion **154** which is disposed adjacent to valve seat **122** and configured to be capable of holding and sealing valve seat **122**. Valve seat holder **128** further has a rod portion **156** extending away from flange portion **154** along centerline **130** through collar center **152** and abutting shaft **126**.

[0035] In the embodiment shown in FIG. 1, shaft **126** extends through spring holder **132**, through its center **136**, through spring **104** and spring holder **134** through its center **138**. Shaft **126** may exit bonnet **144** through its center **146** opposite valve seat holder **128** to allow spring holder **132** or **134** to be adjusted by rotating it. Shaft **126** also may have knob **158** to allow an end user to rotate shaft **126** more easily. Further, knob **158** may have knurls on its exterior, which further facilitate gripping and rotation of knob **158**.

[0036] Bonnet **144** may be secured sealably to exterior surface of relief valve body **108** using convention means. In FIG. 1, bonnet **144** has a flange **160**, and bonnet **144** is secured to exterior

surface **108** with seal **112** in between bonnet flange **160** and exterior surface **108** using screws **110**. In addition, movable shaft **126** preferably has a seal between it and bonnet **144** at its center **146** where it exits so that fluid cannot flow between shaft **126** and bonnet center **146**. Similarly, collar **150** should have a seal between it and bonnet **144** and rod portion **156** so that fluid cannot flow through flow limiter **124** unless pressure relief valve **100** has been activated.

[0037] Relief valve **100** may be made using convention plumbing manufacturing methods. Relief valve **100** is used by installing it in a building plumbing system either by soldering it into a plumbing line, or by manufacturing it with threaded inlet **118** and outlet **120** and using threaded couplings to secure it in fluid communication with the building plumbing system.

[0038] Relief valve **100** inlet **118** receives fluid from a hot-or cold-water source in a domestic plumbing system. From inlet **118**, the fluid can flow through passageway **116** of body **108** and valve seat **122** if the pressure of the fluid is greater than a predetermined pressure. The predetermined pressure is the force required to activate or lift valve seat holder **128** off valve seat **122** to deliver fluid through valve seat **122** to outlet **120** and the delivery end of a faucet, other valve, or other fixture in the building plumbing system.

[0039] Flow limiter **124** restricts the flow of fluid through relief valve **100** so that the pressure of the fluid is relieved when the fluid in the line freezes but will flow at less than the rate of flow through an open valve in the building plumbing system. Thus, flow limiter **124** protects the surroundings in case the flow of fluid has the potential to cause damage.

[0040] Spring **104** has a compressive force such that when divided by the area of the open area of valve seat **122** which is in contact with seat holder **128** is approximately equal to or less than the predetermined pressure. The activation pressure is the compressive pressure required to activate relief valve **100** and overcome the force required to compress spring **104** so that fluid may flow through relief valve **100**. The predetermined pressure can be preset by selecting spring **104** that has a compressive force that will allow it to be compressed at the pressure desired to activate relief valve **100**. The predetermined pressure also can be preset by selecting spring **104** to have a desired compressive force such that when one or both spring holders **132** and **134** are present the activation pressure of relief valve **100** can be preset to be higher than the compressive force of the spring **104** alone. In either case, spring **104** has a force such that it is capable of applying the desired pressure on seat holder **128**, which in turn applies pressure to the valve seat **122** and closes relief valve **100** when the pressure of the fluid in the building plumbing system is less than or equal to the pressure that will burst one or more pipes, valves or fixtures in the building plumbing system, and activates or opens relief valve **100** when the pressure of the fluid in the building plumbing system is greater than or equal to the pressure that will burst one or more pipes, valves or fixtures in the building plumbing system.

[0041] A typical water control valve used in domestic plumbing is estimated to have a break strength of around 4,000 PSI. A pressure that would break a typical copper plumbing pipe is approximately 1,000 PSI, depending on the wall thickness and method used to manufacture the pipe. Plastic (PEX) pipe will typically burst at around 500 PSI. Normal water pressure in a domestic plumbing system is typically less than 100 PSI, and, more typically, around 40 PSI.

[0042] Preferably, spring **104** has a compressive pressure that is less than the pressure at which the weakest component in the plumbing system will burst. The compressive pressure required to compress spring **104** is the force required to compress the spring divided by the area of the valve seat opening which is in contact with the seat holder when the relief valve is closed. If relief valve **100** has at least one spring holder **132** or **134** that is adjustable on shaft **126**, then spring **104** can have a lower compressive pressure than required to activate relief valve **100** if holder **132** or **134** may be adjusted to increase the pressure required to compress the spring as adjusted desired force.

[0043] Spring **104** in relief valve **100** for use in a plumbing system with plastic (PEX) piping preferably has a compressive force such that relief valve **100** will have an activation pressure of less than about 500 PSI. More preferably, spring **104** used in relief valve **100** for a plumbing

system with plastic (PEX) piping has a compressive force such that relief valve **100** has an activation pressure of between about 200 and about 395 PSI, and, even more preferably, between about 300 and about 395 PSI.

[0044] For a building plumbing system that has no plastic PEX piping but rather only copper piping, spring **104** preferably has a compressive force such that relief valve **100** has an activation pressure of less than about 1,000 PSI. More preferably, spring **104** used in relief valve **100** in a plumbing system with only copper piping has a compressive force such that relief valve **100** has an activation pressure of between about 400 and about 995 PSI, and, even more preferably, between about 500 and about 995 PSI.

[0045] In another spring-activated embodiment according to the disclosure, the valve seat sealing portion can be disposed within the passageway generally parallel to the exterior surface of the body, and the valve seat disposed in the passageway generally transverse to the interior and exterior surfaces of the relief valve body, as shown in FIG. 2. Relief valve **200** has a valve seat sealing portion **202** with a spring **204**, and a generally tubular elongate body **206** with an exterior surface **208** and an interior surface **210** defining a passageway **212** that extends through body **206** through which fluid may flow. Valve seat sealing portion **202** is disposed generally parallel to exterior and interior surfaces **208** and **210** within passageway **212**. Relief valve **200** further has an inlet **214** capable of receiving fluid from a source or other plumbing fixture at relatively high pressure, such as that from a domestic plumbing system cold- or hot-water source, and an outlet **216** capable of receiving fluid at a lower pressure from the inlet **214** after the fluid has traveled through passageway **212** and exiting relief valve **200** for delivery to a faucet, other valve or other fixture in a building plumbing system, and a valve seat **218** disposed in between inlet **214** and outlet **216** generally transverse to passageway **212**.

[0046] Valve seat sealing portion **202** has a bonnet **220** attached over valve seat **218** so that fluid flowing through valve seat **218** passes through bonnet **220**. Bonnet **220** may be soldered in place. Or bonnet **220** may have a flange **222** and be attached with a retainer **224** and screws to flange **222**. Preferably, bonnet **220** is sealably attached so that fluid cannot flow between bonnet flange **222** and interior surface **210** using a seal **226** therebetween.

[0047] Bonnet **220** holds spring **204** against a ball **228**, which in turn is forced down against valve seat **218** by spring **204**, thereby sealing relief valve **200**. Ball **228** is generally round for ease of manufacturing but could have other shapes. Valve seat **218** is generally round for receiving a round ball for ease of manufacturing but could have other shapes so long as ball seats in valve seat to seal it. Valve seat **218** and ball **228** are sized to seal with each other and prevent the flow of fluid through valve **200** when ball **228** is forced against valve seat **218** by spring **204**.

[0048] Valve seat **218** may be comprised of the same material of which body **206** is comprised. Valve seat **218** could be made of a material that is different from the material used to make body **206**, like valve seat **122** of relief valve **100**.

[0049] Relief valve **200** also has a flow limiter **230** which allows fluid to flow through relief valve **200** once activated but restricts the flow to less than the normal flow of fluid through a building plumbing pipe. Flow limiter **230** is disposed in bonnet **220** distal to bonnet flange **222**. Flow limiter **230** may have the features and be sized like flow limiter **124** of relief valve **100**.

[0050] The force required to allow fluid to flow through relief valve **200**, in other words its activation pressure, can be preset by selecting spring **204** with a compressive force that will exert the desired pressure on ball **228** to prevent fluid from flowing through valve seat **218**, keeping relief valve **200** closed until the force of the fluid exceeds the compressive force of spring **204**. Spring **204** has a compressive force as described relative to spring **104** and relief valve **100** that is less than the force which would burst the weakest component in the building plumbing system.

[0051] The activation pressure of relief valve **200** could be increased above the compressive force of spring **204** by including spacers within bonnet **220** or using a bonnet **220** sized to hold spring **204** in a compressed state. Doing so would increase the force that spring **204** would apply to ball

228, which would in turn increase the force that would be required to open relief valve **200**.

[0052] Another relief valve embodiment according to the disclosure is shown in FIG. **3**. Relief valve **300** has a body **302** with an exterior surface **304** and an interior surface **306** defining a generally longitudinal passageway **308** through which fluid can flow from an inlet **310** for receiving fluid at a relatively high pressure from a fluid source such as hot-or cold-water or other plumbing fixture to an outlet **312** where fluid exits relief valve **300** at a lower pressure than it enters inlet **310** for delivery to a faucet or other plumbing pipe or fixture, a valve seat **314** disposed on interior surface **306** and a rupture or burst disk **316** disposed over valve seat **314** across passageway **308** such that disk **316** is capable of preventing the flow of fluid through the passageway **308** when the fluid is below a predetermined pressure.

[0053] Rupture disk **316** is comprised of a material having a break strength which bursts at the predetermined pressure. Materials that can be used for rupture disk **316** are metal, plastic, such as polyethylene or polypropylene, or elastomers, such as butyl rubber. Stainless steel or other non-corroding material is preferred. Rupture disk **316** can be flat or concave. It also can have one or more scores or areas that are thinner or configured such that one or more of them break at a pressure that is less than the break strength of the disk material surrounding the one or more scores.

[0054] The material used for the rupture disk **316** without a score preferably has a break strength that is less than the pressure at which the weakest component in the building plumbing system in which relief valve **300** is disposed, as described relative to relief valve **100**. If rupture disk **316** has a score in it, a material having a higher break strength could be used for rupture disk **316** so long as the score breaks at a break strength that is less than the break strength of copper pipe if a plumbing system with only copper pipe is used, or less than the break strength of plastic pipe if the plumbing system has some plastic pipe. In this way, relief valve **300** would remain closed until the force of the fluid in the inlet **310** exceeds rupture disk **316** break strength at which point it would break, relieving the pressure in the line so that other pipes, valves, or fixtures in the plumbing system do not break.

[0055] Rupture disk **316** can be held in place by soldering, screwing, gluing, or using other conventional means to hold it into place. Rupture disk **316** can be held in body **302** by retainer **318**. Alternatively, relief valve **300** can have an inlet end **320** and a separate outlet end **322** each having threads that allow them to be threaded into one another with rupture disk **316** held in between.

[0056] More specifically, as shown in FIG. **3**, relief valve **300** can comprise separate inlet and outlet ends **320** and **322** each with threads **324** and **326**, inlet end threads **324** disposed on the interior of inlet end **320** adjacent to threads **326** on outlet end **322** so that outlet end **322** can be screwed into inlet end **320**. Relief valve **300** has rupture disk **316** disposed in inlet end **320** across passageway **308** with retainer **318** disposed between rupture disk **316** and outlet end **322**, securing rupture disk **316** in place against valve seat **314** by screwing outlet end **322** into inlet end **320**.

[0057] Additionally, rupture disk **316** may have a flange **328** on its outer edge sized to be capable of being held by retainer **318** against valve seat **314**. Flange **328** preferably is molded into rupture disk **316**.

[0058] Alternatively, inlet end **320** and outlet end **322** can be threaded and configured so that inlet end **320** can be received threadably into outlet end **322** with retainer **318** held between outlet end **322** and rupture disk **316**, and rupture disk **316** held against a valve seat inside outlet end **322**.

[0059] Relief valve **300** may also have a flow limiter **330** disposed on interior surface **306** through which fluid can flow in a restricted manner from high-pressure inlet **310** to low-pressure outlet **312** when the pressure of the fluid in inlet **310** is high enough to burst rupture disk **316**. Flow limiter **330** is sized and shaped like flow limiters **124** and **230** of relief valves **100** and **200**, respectively. In FIG. **3**, flow limiter **330** is disposed on the interior of outlet end **322** between outlet **312** and rupture disk **316**.

[0060] Use of a flow limiters **124**, **230** and **330** is preferred because, when relief valves **100**, **200** and **300** open under pressurized conditions, they will prevent a large flow of fluid from flowing

through the pipe which could overflow a tub or sink or other fixture and damage the building in which the relief valve is used.

[0061] One or more relief valves **100**, **200** or **300** could be connected at two points on a pipe without any control valve or other fixture or plumbing component in between but near a control valve or other fixture or component. By doing so, the control valve or other plumbing fixture is protected from freezing. Alternatively, a relief valve **100**, **200** or **300** could be connected to two pipes, one pipe supplying a control valve for a faucet, or other fixture, such as a shower mixing and control valve system, and one pipe connected to the downstream end of the control valve faucet or fixture. In doing so, relief valve **100**, **200** or **300** relieves pressure and protects the pipe, control or other valve, or plumbing system component.

[0062] The predetermined pressure for activation of pressure relief valves **100**, **200** and **300** is preferably less than about 1000 PSI for a building plumbing system with entirely copper piping. More preferably, the predetermined activation pressure used in relief valves **100**, **200** and **300** installed in a building plumbing system with all copper piping will be between about 400 and about 995 PSI, and, even more preferably, between about 500 and about 995 PSI.

[0063] If the building plumbing system includes plastic PEX piping, then the predetermined activation pressure for relief valves **100**, **200** and **300**, preferably, should be less than about 500 PSI. If it is unknown what piping is used in a building plumbing system, then pressure relief valve **100**, **200** or **300** used in the system preferably should have a predetermined activation pressure of less than about 500 PSI, and more preferably between about 200 and about 395 PSI, and, even more preferably, between about 300 and about 395 PSI.

[0064] Spring **104**, **204** and rupture disk **316** preferably have activation pressures of less than the burst pressure of the weakest component in the building plumbing system. When spring **104**, **204** and rupture disk **316** have activation pressures that are less than the weakest component in the building plumbing system, relief valves **100**, **200** and **300** will activate and relieve the pressure in the building plumbing system before a burst results.

Control Valve System with a Relief Valve

[0065] A pipe burst prevention valve system for use in a building plumbing system is described having a control valve and a relief valve **100**, **200** or **300**. For example, as shown in FIG. 4, pipe burst prevention valve system **400** according to the disclosure has a mixing and diverter valve **402** on which a faucet housing with a handle would be placed. FIG. 4 shows valve system **400** having two inlets from two high-pressure sources **404A** and **404B**, typically, cold-and hot-water lines, respectively, in fluid communication with control valves **406A** and **406B**, one for controlling cold water from cold-water line **404A** and one for controlling hot water from hot-water line **404B**, respectively, which each also would have a handle on the faucet housing for turning each control valve **404A** and **404B** on or off, two outlets **408A** and **408B**, and two bypasses **410A** and **410B**, each having a pressure relief valve **412A** and **412B** according to the disclosure, such as one of pressure relief valve **100**, **200** or **300**, an inlet **414A** and **414B**, with inlet **414A** in fluid communication with control valve inlet **404A** and inlet **414B** in fluid communication with control valve inlet **404B**, and an outlet **416A** and **416B**, with outlets **416A** and **416B** in fluid communication with outlet **408B**.

[0066] Typically, for a valve system like that shown in FIG. 4, one outlet **408A** would deliver water to a bathtub faucet, and the other outlet **408B** would deliver water to a shower head, either at the same time or separately, depending upon mixing and diverting valve **402** and the position of the handle controlling it. Bypasses **410A** and **410B** will allow water to flow through them when the pressure in inlets **404A** and **404B** exceeds the force required to activate relief valve **412A** or **412B** thereby bypassing control valves **406A** and **406B** and mixing valve **402** and protecting them and the pipes in the building plumbing system from freezing. Bypass **410A** or **410B** could be in fluid communication with outlet **408A** instead of **408B**, or one could be connected to outlet **408A**, and the other bypass could be connected to outlet **408B**.

[0067] Spindles **418A** and **418B** are for attaching cold-and hot-water handles to control valves **406A** and **406B**, respectively, so that the valves can be operated. Spindle **420** is for attaching a mixing and diverting valve handle to the valve **402** to operate it.

[0068] Often, only the cold-water control valve inlet pipe is at risk of freezing in a domestic plumbing system, and not the hot-water control valve pipe inlet. In this case, a valve system could have only one bypass disposed to bypass the cold-water control valve and protect it from freezing. However, where there is a risk that the building plumbing system's hot water heating system could fail, or where the building heat could fail or both and there is a risk that the pipes will freeze, then a valve system with more than one bypass is preferred.

[0069] Preferably, valve system **400** has spring-operated relief valve embodiments **100** and **200** as relief valves **412A** and **412B** in bypasses **410A** and **410B** because they will continue to protect a building plumbing system after a freezing event. Once a frozen pipe thaws and the pressure of the building plumbing system returns to normal, relief valves **100** and **200** close and water no longer would flow through them, but they will open again if the pipes freeze again. Relief valve **300**, however, is preferred where space and cost are limiting factors.

[0070] Bypasses **410A** and **410B** according to the disclosure can also have relief valves **100**, **200** or **300** within a Y-type access pipe having an access port which would allow the relief valves **100**, **200** or **300** to be inspected and serviced with minimal downtime for the plumbing system. Relief valve **100**, **200** or **300** needing to be replaced could be removed or replaced using the access port. This is especially important for relief valve **300** which will no longer function to relieve pressure in the pipe once its rupture disk **316** has burst. Therefore, relief valve **300**, once ruptured, will need to be replaced to permit valve system **400** to operate as described and to continue to protect the plumbing system or its pipes or other components from freezing.

[0071] FIG. 5 is a flow chart showing the flow of fluid through a valve system **400**, shown in FIG. 4. In FIG. 5, fluid flows through the pipe burst prevention valve system **400** from two inlets from two high-pressure fluid sources **404A** and **404B**, typically, cold-and hot-water lines, respectively, in fluid communication with control valves **406A** and **406B**, one for controlling cold water from cold-water line **404A** and one for controlling hot water from hot-water line **404B**, respectively, to outlets **408A** and **408B**, depending upon the position of the mixing and diverter valve **402**. Fluid flows through bypasses **410A** and **410B**, each having a pressure relief valve **412A** and **412B** according to the disclosure such as one of pressure relief valve **100**, **200** or **300**, when the pressure of the fluid in inlets **404A** or **404B** is at or above the activation force of relief valve **412A** or **412B** thereby bypassing control valves **406A** and **406B** and mixing valve **402** and protecting them and the pipes connecting them to the building plumbing system from freezing by allowing fluid to flow through outlet **408A**. In FIG. 5, outlet **408A** could be a bathtub faucet and outlet **408B** could be a shower faucet.

[0072] FIG. 6 is a perspective view of an alternative embodiment of a valve system according to the disclosure. Valve system **500** shown in FIG. 6 comprises two control valves for controlling the flow of fluid **502A** and **502B** from an inlet from a high-pressure cold-water source **504A** and from a high-pressure hot-water source **504B**, respectively, and each such control valve **502A** and **502B** having outlets **506A** and **506B**, respectively, which combine into one control valve system outlet **508**, such as a faucet for a bathtub or shower. Valve system **500** further has two bypasses **510A** and **510B** each having a pressure relief valve **512A** and **512B** according to the disclosure, respectively, such as one of embodiments **100**, **200**, or **300** shown in FIG. 1, 2 or 3, disposed in fluid communication with control valve inlets **504A** and **504B** and control valve outlets **506A** and **506B**. System **500** also has spindles **514A** and **514B** configured to receive handles to aid in turning control valves **502A** and **502B** on and off.

[0073] FIG. 7 is a flow diagram showing the flow of fluid through valve system **500**, which has control valves **502A** and **502B** for controlling the flow of fluid from high-pressure inlets **504A** and **504B**, respectively, and flowing out to a combined control valve system outlet **508**, and two

bypasses **510A** and **510B** disposed in fluid communication with inlets **504A** and **504B**, respectively, and control valve system outlet **508**, each bypass **510A** and **510B** having a relief valve **512A** or **512B**, according to the disclosure. System **500** could have only one bypass, preferably on the cold-water source side of system **500**, instead of two as shown.

[0074] Another embodiment according to the disclosure is a control valve system that includes another low-pressure delivery pipe end, not shown in FIG. **6** or **7**. In yet another embodiment according to the disclosure a diverter that may be operated between one or more deliver end pipes, such as where a shower has multiple shower heads, which diverts the water from a first low-pressure delivery pipe to a second low-pressure delivery pipe or more low-pressure delivery pipes, and back again. For example, such a diverting valve could be advantageously used to divert water being delivered from one location to another or to two or more locations, such as a bathtub to a shower or multiple shower heads.

[0075] In yet another embodiment, the system can include a mixing valve that mixes and moderates hot-and cold-water delivered to the mixing valve. Such a mixing valve also can be pressure controlled. Such a mixing valve can also have a diverting function. Such a mixing valve also may not have a diverting function and be used to deliver water to a diverter valve.

[0076] FIG. **8** is an exploded perspective view of another embodiment of a control valve system according to the disclosure. In FIG. **8**, valve system **600** has a cartridge **602** that receives fluid from fluid sources through inlets **604A** and **604B** and mixes them as well as controls their volume and allows the fluid to exit from a single outlet **606**. Valve system **600** also has two bypasses, **608A** and **608B**, in fluid communication with inlets **604A** and **604B**, respectively, and control valve system outlet **606**. Valve system **600** also is shown having an aerator **610**. Valve system **600** has faucet housing **612** which is disposed over cartridge **602** and out of which fluid flows through outlet **606** and has a handle **614** for controlling fluid flow from inlets **604A** and **604B**, center lines of which are shown in dashed lines, through cartridge **602**. Bypasses **608A** and **608B** have a relief valve according to the disclosure, such as relief valve **100**, **200** or **300**.

[0077] In addition, a valve system according to the disclosure could include additional outlets, additional mixing, control valves, such as thermostatic and pressure control valves, as well as a bypass with a pressure relief valve according to the disclosure, such as relief valve **100**, **200** or **300**. Further, valve system **400**, **500**, and **600** could also include in bypasses **410A**, **410B**, **510A**, **510B**, **608A** and **608B** a Y-type access port disposed to allow relief valves **412A**, **412B**, **512A**, **512B**, **608A**, and **608B** to be serviced.

[0078] Valve systems **400**, **500**, and **600** can be made by soldering, screwing, or clamping the components described in relation to them to pipes in fluid communication with them. Alternatively, they can be made by molding the valves and bypasses into an assembly and attaching the assembly using fittings or soldering the assembly to pipes in fluid communication with the building plumbing system. Relief valves and cartridges may be made removable and replaceable by threading or using O-rings. Preferably, a valve system according to the disclosure is molded such that bypasses are integral and generally parallel to the valve that they bypass so that they are compact enough to be installed in a two-by-four wall, typical in residential dwelling walls, like conventional shower or tub wall mixing, diverter and control valves can be installed, yet the valve system shall be molded such that the relief valve and any cartridge shall be removable so that they can be replaced when necessary. The valve system according to the disclosure advantageously can also be installed in a wall on the exterior of a residential dwelling because it is protected against freezing.

Control Valve Cartridge with a Split Stem and a Washer

[0079] Pipe bursts due to freezing also can be prevented by using a compact, pipe-burst-prevention fluid control valve cartridge according to the disclosure for use with a fluid control valve faucet for controlling the flow of fluid having a pressure in a plumbing system. One embodiment of such a cartridge is a generally cylindrical control valve cartridge **700**, as shown in FIG. **9**, having a washer

702 and a split stem **704**. More particularly, cartridge **700** has a stem end **706** opposite a valve end **708**, a cartridge body **710** that has exterior and interior surfaces **712** and **714**, respectively, with upper and lower interior threads **716** and **718**, upper interior threads **716** toward cartridge upper stem end **706** and lower interior threads **718** toward cartridge valve end **708**, and cartridge body **710** further has exterior threads **720** toward cartridge valve end **708** configured to be capable of being screwed into faucet housing to hold cartridge **700** in place in the faucet housing. Cartridge **700** has split stem **704** having an upper stem **722** with a generally cylindrical exterior surface **724** with threads **726** thereon adjacent to cartridge interior surface lower end threads **718**, and a shoulder **766** formed between stem threads **726** extending beyond upper stem exterior surface **724**. [0080] Cartridge **700** further can have a packing **730** disposed around upper stem exterior surface **724** on upper stem thread shoulder **766**, and a packing nut **732** having an interior surface **768** configured to be capable of rotatably receiving upper stem **722**, threads on its exterior surface **734** configured to be capable of being threaded into upper interior cartridge body threads **716**, and a lower end **770** capable of resting on stem thread shoulder **766** when the spindle **740** is turned to its fully open position. Cartridge **700** further can have a stop **772** disposed on shoulder **766** of upper stem threads **726** and, in this case as shown in FIG. 9, packing **730** is disposed on stop **772** and packing nut **732** is configured to screw into body **710** with packing nut lower end **770** abutting packing **730**, and packing **730** abutting stop **772**.

[0081] Upper stem **722** has an upper end **736** toward the cartridge stem end **706** and a lower end **738** toward the valve end **708** of the cartridge **700**. The upper stem **722** has a spindle **740** on which a handle for a faucet may be attached. Upper stem **722** also has a generally cylindrical interior cavity **742** extending from the upper stem lower end **738** toward the upper stem upper end **736**, a calibrated spring **744** disposed in cavity **742** toward the upper stem upper end **736**, and a clip **762** attached to upper stem lower end **738** adjacent to cavity **742**. Preferably, upper stem lower end **738** has at least two clips **762** opposing one another on opposite sides of cavity **742**.

[0082] Cartridge split stem **704** also has a lower stem **728** toward cartridge valve end **708** with generally cylindrical upper end and lower ends **746** and **748**, respectively, the lower end **748** having a diameter that is larger than the diameter of upper end **746** thereby forming a shoulder **750**. Lower stem upper end **746** has a diameter that is sized so that it is capable of sliding into and being received by cartridge body interior **714** at its valve end **708**. Lower stem **728** also has a piston **752** having an upper end **754**, a lower end **756** attached to the upper end of the lower stem **746** and having a flange **758** adjacent to piston lower end **756**, and a retaining screw **760** attaching washer **702** to the lower stem lower end **748**. Piston **752** can be formed integrally with the lower stem lower end **748** or it can be screwed or soldered onto it. Alternatively, two or more of lower stem upper end **746**, lower stem lower end **748** and piston **752** could be molded together as one piece.

[0083] Piston **752** has a size such that it is capable of slidably being received within cavity **742** and holding a calibrated spring **744** in cavity **742** at its upper end **754**. Piston **752** is capable of sliding between a fully extended position toward valve end **708**, held by a piston retaining clip **762** on flange **758** and thereby defining a gap **764** between lower stem upper end **746** and upper stem lower end **738**, and fully compressed position, wherein lower stem **728** is compressed upward toward stem end **706** until shoulder **750** rests on cartridge body **710** at its valve end **708**.

[0084] Cartridge **700** can be used in single-spindle compression washer control valves, like valves **406A** and **406B**, shown in FIG. 4, and valves **502A** and **502B**, shown in FIG. 6. Cartridge **700** is used by inserting its valve end **708** into a control valve of a faucet and screwing it into the faucet so that washer **702** and valve end **708** rest on the valve seat of the faucet. Cartridge **700** may not have exterior threads **720** which screw into the valve but rather may have an O-ring that friction fits into the faucet valve. When faucet is opened and closed by turning the handle of the faucet attached to spindle **740**, upper stem **722** rotates freely within packing nut **732** and moves in threads **726** toward cartridge stem end **706** or valve end **708** on interior threads on lower valve end **718**, which raises or lowers valve end **708** of the split stem **704** off or onto valve seat to allow or prevent the flow of

fluid through the valve.

[0085] Piston **752** may be configured to rotate with upper stem **722** when spindle **740** is rotated to open or close the valve yet slide longitudinally relative to upper stem **722**. In that case, piston **752** would have an elongate protrusion or key on the exterior of piston **752** extending radially outward toward and longitudinally down a portion of the length of piston **752**, and cavity **742** would have a complementary recess in it that is configured to longitudinally, slidably receive the piston key and hold it fixed relative to upper stem **722** when it is rotated thereby rotating lower stem **728** and washer **702** when spindle **740** is rotated.

[0086] When faucet handle on spindle **740** is turned to open the valve, fluid flows from the building plumbing system past washer **702** and valve seat. In FIG. **4**, when spindle **418A** or **418B** are turned, washer (not shown) raises, and fluid flows past valves **406A** or **406B** and out faucet outlet **408A** or **408B** depending upon the position of mixing valve **420**. In FIG. **6**, when spindle **514A** or **514B** is turned, washer (not shown) would raise, and fluid would flow past valve **502A** or **502B** and out faucet outlet **508**. In each case, where cartridges disclosed are included in a valve system of a building plumbing system, opening the valve by turning the spindle would allow water to flow from the building plumbing system water source through outlet to be used by an end user. In each case, closing the valve by turning the spindle prevents water from flowing from the building plumbing system water source through outlet to an end user under normal fluid pressure conditions but, under freezing conditions, fluid pressure increases above the predetermined pressure and water will flow to an outlet.

[0087] FIG. **10A** shows cartridge **700** under normal fluid pressure conditions in a valve in its fully open position. Under these conditions, as shown in FIG. **10A**, when spindle **740** has been turned to open the valve and raise upper stem **722** and washer **702** off faucet valve seat (represented by dashed line), shoulder **750** rests on cartridge body **710**. In this position, fluid can flow through the valve past washer **702**.

[0088] FIG. **10B** shows cartridge **700** under normal fluid pressure conditions in a valve with cartridge **700** in its fully closed position. Under these conditions, as shown in FIG. **10B**, when spindle **740** has been turned to close the valve and lower upper stem **722**, washer **702** rests on faucet valve seat (again represented by dashed line). In this position, fluid is prevented from flowing through the valve and past washer **702**.

[0089] FIG. **10C** shows cartridge **700** with its stem **722** in its fully closed position but under pressurized fluid conditions. Under these conditions, as shown in FIG. **10C**, spindle **740** has been turned to its closed position which advances upper stem **722** in cartridge body too its extended position off of but lower stem **728** and washer **702** are pushed up off of the faucet valve seat (shown by dotted line) by pressurized fluid which exceeds the compressive force of spring **744**, and activation pressure or predetermined pressure of cartridge **700**, in turn pushing up piston **752** into cavity **742** allowing fluid to flow in the pipe past cartridge valve end **708** and valve seat, and through the valve.

[0090] Cartridge **700** is configured, and the lengths of cavity **742**, spring **744** and piston **752** are sized so that when spindle **740** is in its fully closed position but the valve is under pressurized conditions, the fluid pressure exceeds the predetermined pressure, and lower stem lower end **748** raises off valve seat so as to allow at least some fluid to flow through the valve. Preferably, cartridge **700** is configured and the lengths of cavity **742**, spring **744** and piston **752** are sized so that when spindle **740** is in its fully closed position but fluid pressure in the valve exceeds the predetermined pressure, lower stem lower end **748** raises off valve seat by an amount that is less than it raises off the valve seat when spindle **740** is turned to fully open the valve. More preferably, cartridge **700** is configured and the lengths of cavity **742**, spring **744** and piston **752** are such that when spindle **740** is in its fully closed position but fluid pressure in the valve exceeds the predetermined pressure, lower stem lower end **748** raises off valve seat by a distance that allows between about 50% to about 80% less than the normal flow of fluid through the valve when the

valve is open, or by a distance that allows between about 0.25 and about 1.25 GPM and about 0.10 and about 0.50 GPM to flow through the valve. Even more preferably, cartridge **700** is configured and the lengths of cavity **742**, spring **744** and piston **752** are such that when spindle **740** is in its fully closed position but fluid pressure in the valve exceeds the predetermined pressure, lower stem lower end **748** raises off valve seat by a distance that allows an amount of fluid to flow through the valve that is less than about 0.4 GPM.

[0091] Spring **744** has a compressive force so that cartridge **700** is capable of relieving pressure in the building plumbing system at the desired predetermined pressure. Thus, spring **744** is capable of compressing when the pressure of the fluid in cartridge **700** exceeds predetermined pressure.

Spring **744** should have a predetermined activation pressure as described above according to the type of pipes and components in the building plumbing system.

Control Valve Cartridge with a Split Stem and Ceramic Disks

[0092] Another cartridge embodiment according to the disclosure is shown in FIG. **11**. FIG. **11** depicts a generally cylindrical control valve cartridge **800** for a single spindle control valve having a valve end **802** and a stem end **804** generally opposite each other along a longitudinal axis (shown in dashed line) through the center of cartridge **800**, a cartridge base **806** at valve end **802**, and a split stem **808**. Cartridge **800** also has a body **810** that has an exterior surface **812** with an O-ring **814** on it, cartridge body exterior surface **812** being shaped to receive O-ring **814** and friction fit into a faucet valve, and an interior surface **816** that is sized to hold split stem **808** with the assistance of two O-rings **818** and **820** for holding and sealing split stem **808** to the cartridge body interior surface **816**.

[0093] Split stem **808** has an upper stem **822** with a generally cylindrical exterior surface **824** with recesses **826** and **828** each sized to receive O-ring **818** and **820** and hold and seal upper stem **822** in cartridge body **810**. Split stem **808** also has a lower stem **830** toward valve end **802** adjacent to upper stem **822**.

[0094] Upper stem **822** has an upper end **832** adjacent to the cartridge stem end **804**, a lower end **834** distal to upper stem upper end **832**, and a spindle **836** on which a handle for a faucet may be attached and used to rotate spindle **836** and split stem **808** to open the valve and turn the faucet on or off. Upper stem **822** also has interior cavity **838** extending longitudinally through upper stem **822** from its lower end **834** toward upper stem upper end **832**.

[0095] Lower stem **830** has an upper end **840** adjacent to upper stem lower end **834** and a lower end **842** having a ceramic control disk **844**. Lower stem **830** also has piston **846** disposed on lower stem upper end **840** and extending longitudinally into cavity **838** toward stem end **804**.

[0096] Piston **846** can be formed integrally with lower stem upper end **840**. Alternatively, piston **846** can be screwed or soldered on to lower stem upper end **840** to affix it to lower stem **830**.

[0097] Cartridge **800** also has pressure relief spring **848** disposed in cavity **838** and held in place by piston **846**. Spring **848** has a compressive force that allows cartridge **800** to relieve pressure in the building plumbing system at the desired predetermined pressure. Thus, spring **848** is capable of compressing when the pressure of the fluid in cartridge **800** exceeds the predetermined pressure. Spring **848** has a predetermined activation pressure as described above according to the type of pipes and components in the building plumbing system.

[0098] Cartridge base **806** is fixedly disposed on interior surface of cartridge body **816** at its valve end **802** across a high-pressure inlet **850** and has at least one opening through which fluid may flow from a high-pressure source, such as a hot-or cold-water source. Cartridge base **806** holds a ceramic cartridge base **854** in cartridge body **810** at its valve end **802**. Ceramic cartridge base **854** also has at least one opening through which fluid from opening in base **806** may flow.

[0099] Ceramic control disk **844** is disposed on lower stem **830** at its lower end **842** and forced toward ceramic cartridge base **854** by piston **846** under the force of spring **848**. Ceramic control disk **844** has at least one opening configured to allow fluid to flow from inlet **850** through opening in ceramic cartridge base **854** if ceramic control disk **844** is rotated to align its opening with low-

pressure outlet **860** and to prevent the flow of fluid from inlet **850** to outlet **850** if ceramic control disk **844** is rotated the opposite way thereby misaligning opening in control disk **844** with outlet **860**.

[0100] Cartridge **800** further has retaining ring **862** that rotatably holds upper stem **822** in cartridge body **810** at its stem end **804** thereby holding spring **848** and piston **846** in cavity **838**, and lower stem **830** against ceramic cartridge base **854**. Ceramic cartridge base **854** in turn is fixedly held against cartridge base **806** in cartridge body **810** by conventional means. Retaining ring **862** and cartridge base **806** may be held in place by screwing them into place, by friction fitting them into place using O-rings, or by any other conventional means.

[0101] Piston **846**, cavity **838** and pressure relief spring **848** have diameters such that piston **846** and spring **848** can slide into cavity **838**. Piston **846**, cavity **838** and spring **848** have lengths such that a portion of piston **846** is capable of sliding into cavity **838** and holding spring **848** therein between a fully extended position, wherein lower stem **830** is held by ceramic cartridge base **854** against the force of spring **848** under normal fluid pressure conditions and thereby defining a stem gap **864** in split stem **808** between upper stem **822** and lower stem **830**, and a fully retracted position, when the pressure of the fluid at the valve end exceeds the predetermined pressure thereby compressing spring **848**, closing stem gap **864**, and defining a ceramic disk gap **866** between ceramic control disk **844** and ceramic cartridge base **854** when the pressurized fluid forces control disk **844** upward forcing lower stem **830** toward upper stem lower surface **834**, closing stem gap **864**, defining ceramic disk cap **866**, and allowing an amount of fluid to flow out of cartridge **800** through low pressure outlet **860**.

[0102] Preferably, cartridge **800** is configured and the lengths of piston **846**, cavity **838**, pressure relief spring **848**, upper end of lower stem **840** and ceramic control disk **844** are sized so that when the fluid pressure exceeds the predetermined pressure, lower stem lower end **842** raises off of ceramic base **854** so as to allow an amount of fluid to flow from cartridge inlet **850** to cartridge outlet **860** that is less than that the amount of fluid that would flow through the outlet **860** when spindle **836** is in its fully open position and the valve is fully open. More preferably, cartridge **800** is configured and the lengths of piston **846**, cavity **838**, pressure relief spring **848**, upper end of lower stem **840** and ceramic control disk **844** are sized so that when the fluid pressure in the valve exceeds the predetermined pressure, lower stem lower end **842** raises off of ceramic base **854** by an amount that restricts the flow of fluid through outlet **860** to an amount that is between 50% and about 80% less than the normal flow of fluid through the valve when the valve is in its fully open position, or by a distance that allows between about 0.25 and about 1.25 GPM and about 0.10 and about 0.50 GPM to flow through the valve. Even more preferably, cartridge **800** is configured and the lengths of piston **846**, cavity **838**, pressure relief spring **848**, upper end of lower stem **840** and ceramic control disk **844** are sized so that when the fluid pressure exceeds the predetermined pressure, lower stem lower end **842** raises off of ceramic cartridge base **854** by a distance that allows an amount of fluid to flow from cartridge inlet **850** to cartridge outlet **860** that is less than about 0.40 GPM. Additionally, the length of lower stem **830** can be sized to define a larger or small ceramic disk gap **866**, which in turn also allows or restricts more or less of the fluid flow through low-pressure outlet **860**.

[0103] FIG. 12A shows cartridge **800** in its closed position and under normal fluid pressure conditions. Under these conditions, as FIG. 12A shows, piston **846** is in its fully extended position whereby the force of spring **848** forces piston **846** down, which in turn forces lower stem **830** down, until ceramic control disk **844** rests on ceramic cartridge base **854** and prevents fluid from flowing between ceramic cartridge base **854** and ceramic control disk **844** and out of outlet **860**.

[0104] FIG. 12B shows cartridge **800** under pressurized fluid conditions. Under these conditions, as FIG. 12B shows, piston **846** is in a retracted position whereby the force of the pressurized fluid overcomes the force of spring **848**, compressing it and forcing piston **846** up, which in turn forces lower stem **830** up thereby defining ceramic disk gap **866** between ceramic cartridge base **854** and

ceramic control disk **844**, at least a portion of ceramic disk gap **866** aligns with low-pressure outlet **860** such that an amount of fluid flows out outlet **860** when the spindle **836** has positioned ceramic control disk **844** so that its opening does not align with opening **860** and under normal conditions fluid would not flow through the valve.

[0105] Cartridge **800** can be used in single-spindle ceramic control valve faucets. Cartridge **800** is used by inserting its valve end **802** into a control valve of faucet housing, such as a faucet like that shown in FIGS. **17**, **26** and **28** of U.S. Pat. No. 10,260,642. Cartridge **800** may be held in faucet housing by any means and need not have exterior O-ring **814** holding the cartridge **800** in faucet housing but could have threads or be held therein by other conventional means.

[0106] FIG. **13** is a perspective view of lower stem **830** of cartridge **800** according to the disclosure. Piston **846** is fixed to lower stem upper end **840** and configured to rotate with upper stem **822** when spindle **836** is rotated to open or close the valve yet slide longitudinally relative to upper stem **822** so that it may be inserted into upper stem **822** and extend and retract under the force of spring **848** and pressurized fluid. Specifically, as shown in FIG. **13**, piston **846** has an elongate protrusion or key on the exterior of piston **868** extending radially outward toward and longitudinally down at least a portion of the length of piston **846**. Upper stem cavity **838** has a complementary recess in it that is configured to longitudinally, slidably receive the piston key **868** and hold it so that lower stem **830** rotates when upper stem **822** is rotated when spindle **836** is turned to open and close the valve by rotating ceramic control disk **844** to align or misalign its opening **858** with opening **860** and allow or prevent fluid from flowing from inlet **850** through cartridge base **806**, ceramic cartridge base **854**, opening **858** of ceramic control disk **844** to outlet **860**.

[0107] Piston **752** and **846** need not be solid cylinders and upper stems **722** and **822** need not have cylindrical cavities **742** and **838**. Rather, pistons **752** and **846** could be tubular in shape and stems **722** and **822** could have solid cores which slide into tubularly shaped pistons and have cylindrical slots in stems **722** and **822** shaped to receive them. Alternatively, pistons **752** and **846** could be rectangular in shape with complementarily shaped cavities **724** and **838**. Further, multiple pistons and springs could be used with complementary cavities. Even further, upper stems **722** and **822** could have pistons with lower stems **728** and **830** having complementary cavities. Additionally, multiple pistons and cavities could be used, if desired, for greater alignment and control.

Pipe-Burst-Prevention Washerless Cartridge

[0108] FIG. **14** depicts another embodiment of a valve system according to the disclosure. A single-handle washerless valve system **900** is shown having a cartridge **902** having a control stem **904** operably connected to a cartridge body **906** having hot and cold high pressure fluid supply openings **908** and **910** and a low-pressure fluid delivery opening **912** and over which a cap **914** and handle **916** are disposed, the handle **916** being held in place by a set screw **918** which is screwed into a hole in control stem **904**. Other means could be used to attach handle **916**. The cartridge openings **908**, **910** and **912** correspond to openings **920**, **922** (not shown) and **924** in faucet **926**, and seals **928**, **930** and **932**, respectively. Cartridge body **906** is disposed in recess **934** within faucet **926** with openings **908**, **910**, and **912** aligned with openings **920**, **922** (not shown), and **924**, with seals **928**, **930**, and **932**, respectively, held therebetween. Calibrated spring **936** is disposed on cartridge body **906** and held in place by cap **914** screwed into hole in stem **904**. Thus, set screw **918** screwed into handle **916** and control stem **904** with cap **914** on spring **936** which in turn is disposed on cartridge **906** with washers **932**, **928** and **930** therebetween in recess **934** on faucet **926** aligned as described not only holds handle **916** in place but also cap **914** with spring **936** on cartridge **906** and washers **932**, **928** and **930** therebetween in recess **934** on faucet **926**.

[0109] Handle **916** is rotated right or left to adjust the temperature of the fluid flowing out of faucet **926** which aligns or misaligns openings **908** and **910** with **920** and **922** to allow more or less of hot or cold fluid to flow, thereby controlling the amount that hot **920** or cold **922** fluid supplied and the temperature of the fluid flowing out of outlet **924**. Handle **916** is lifted or lowered to control the

amount of fluid flowing out of valve system **900** causing it to raise cartridge body **906** up proportionately to the distance handle **916** is raised off of seals **928**, **930** and **932**, and allowing water to flow through high-pressure faucet openings **920**, **922** and out delivery opening **924**.

[0110] Cartridge **902** is configured and the lengths of control stem **904**, cartridge body **906**, cap **914** and spring **936** are sized to allow cartridge body **906** to raise off faucet **926** when the pressure of the fluid in the faucet **926** is greater than the predetermined pressure of the spring **936** so that spring **936** is compressed and an amount of fluid flows through openings **908**, **910**, **912**, **920**, **922** and **924**. Spring **936** applies a force on cartridge body **906** and in turn seals **928**, **930** and **932** on openings **920**, **922** and **924** to keep valve **900** closed under normal fluid pressure conditions. When the fluid pressure rises above a predetermined pressure which is the compressive force required to compress spring **936**, the fluid pressure forces cartridge body **906** upward allowing fluid to flow from high-pressure faucet inlet openings **920** and **922** to low-pressure outlet opening **924**.

[0111] Preferably, cartridge **902** is configured and the lengths of control stem **904**, cartridge body **906**, cap **914** and spring **936** are sized to allow cartridge body **906** to raise off faucet openings **920**, **922** and **924** when the pressure of the fluid is greater than the predetermined pressure of the spring **936** to allow at an amount of fluid to flow from faucet inlets **920** and **922** to faucet outlet **924** that is less than the amount of fluid that would flow through outlet **924** when handle **916** is in its fully open position. More preferably, cartridge **902** is configured and the lengths of control stem **904**, cartridge **906**, cap **914** and spring **936** are sized to allow cartridge **902** to raise off faucet **926** when the fluid pressure exceeds the predetermined pressure, and allow an amount of fluid to flow through the faucet **926** by an amount that is between about 50% and about 80% less than the normal flow of fluid through the valve system **900** when the valve is in its fully open position, or by a distance that allows between about 0.25 and about 1.25 GPM and about 0.10 and about 0.50 GPM to flow through the valve. Even more preferably, cartridge **902** is configured and the lengths of control stem **904**, cartridge body **906**, cap **914** and spring **936** are sized to allow cartridge body **906** to raise off faucet **926**, when handle **916** is in its closed position and the fluid pressure in faucet **926** exceeds the predetermined pressure, by a distance that allows an amount of fluid to flow through faucet openings **920**, **922** and **924** that is less than about 0.40 GPM.

[0112] Spring **936** has a compressive force that allows valve system **900** to relieve pressure in the building plumbing system at the desired predetermined pressure. Thus, spring **936** is capable of compressing when the pressure of the fluid in cartridge **902** exceeds the predetermined pressure. Spring **936** has a predetermined activation pressure as described above according to the type of pipes and components in the building plumbing system.

[0113] Relief valves **100**, **200** and **300**, valve system **400**, **500**, **600** and **900**, and cartridge **700**, **800**, and **902** may be made of using any material used to make conventional valves, valve systems and cartridges, including metals and metal alloys (e.g., brass, nickel, titanium, or stainless steel), polymeric or hard rubber materials (e.g., polystyrene-butadiene-styrenes ("SBS"), polyacrylonitrile-butadiene-styrenes ("ABS")), and any material disclosed in U.S. Pat. No. 10,260,642, which is hereby incorporated by reference in its entirety, and by any method used to make conventional valves, valve systems and cartridges, including those described in the '642 patent or U.S. Pat. Nos. 7,475,703, and 3,810,602, which are also hereby incorporated by reference.

Method of Preventing Pipe Bursts in a Building Plumbing System

[0114] A method of preventing pipe bursts due to freezing comprises the steps of disposing a pressure relief valve according to the disclosure, such as relief valves **100**, **200** or **300**, in fluid communication with a building plumbing system. Preferably, relief valve **100**, **200** or **300** is disposed between a fluid source and a control valve or other fixture in the pipe for delivery water to an end user. More preferably, the method comprises disposing relief valve **100**, **200** or **300** behind the same wall in which a control valve or other fixture in the building plumbing system is disposed.

[0115] A further method for preventing pipe bursts due to freezing can comprise disposing a pipe burst prevention valve system in a building plumbing system comprising a faucet housing having

handle and outlet, control valve having inlet, and bypass having a relief valve according to the disclosure, such as relief valve **100**, **200** or **300**, disposed in fluid communication between one inlet and one outlet. Another method for preventing pipe bursts due to freezing can comprise disposing a pipe burst prevention valve system in a building plumbing system comprising a faucet housing having a handle and an outlet, a control valve having two inlets and a mixing valve cartridge, and a bypass having a relief valve according to the disclosure, such as relief valve **100**, **200** or **300**, disposed in fluid communication between one inlet and the outlet.

[0116] A method of preventing pipe bursts due to freezing in a building plumbing system further comprises the steps of disposing a fluid control valve system according to the disclosure in a building plumbing system. Fluid control valve system can be faucet valve system shown in FIGS. **17**, **26** and **28** of U.S. Pat. No. 10,260,642, or FIG. **4**, **6**, or **8** herein with cartridge **700** and **800**, or FIG. **14** with valve system **900** and cartridge **902**. More specifically, a method for preventing pipe bursts due to freezing can comprise disposing a pipe burst prevention valve system in a building plumbing system that has a faucet housing having handle and outlet, control valve **406A**, **406B**, **502A**, **502B**, or **900** according to the disclosure having an inlet, and a cartridge **700**, **800** or **902** according to the disclosure disposed in fluid communication between one inlet and one outlet.

[0117] “Fluid” and “water” as used herein are understood to be any liquid that can freeze. For a domestic plumbing system, the fluid typically is 100% pure water with associated minerals and salt and other additives or naturally occurring substances. However, fluid or water as used herein could be any liquid that has the potential to freeze in a building plumbing system when the system reaches the temperature at which the fluid freezes, such as any aqueous liquid having a sufficient amount of water in it to freeze under the building conditions.

[0118] It must be understood that despite the use of the singular “a” or “an” in a claim they cover more than one claimed element. For example, a claim to the relief valve according to the disclosure covers use of more than one relief valve according to the disclosure or a relief valve according to the disclosure and a cartridge or other fixture in a building plumbing system. Similarly, by way of another example, a claim to a cartridge according to the disclosure covers use of more than one cartridge according to the disclosure and a cartridge and a relief valve according to the disclosure.

[0119] It also will be understood that the claims are not limited to the embodiments described herein. The features of the pressure relief valve, cartridge, control valve system, and methods described are exemplary and may be altered, omitted, or adapted according to the requirements of the plumbing system function and features desired. It will also be understood that the embodiments described include the description herein as well as combinations and sub-combinations of any of the features and equivalents thereof, and any such combinations set forth in the original claims or any amendments thereto.

Claims

1. A relief valve for pipes or valves or other fixtures in a building plumbing system, comprising: a body having an inlet, an outlet, an exterior surface, and an interior surface that defines a generally longitudinal passageway that extends from the inlet to the outlet and is capable of allowing fluid to flow from inlet to outlet; a valve seat disposed on the interior surface of said body between the inlet and the outlet; a flow limiter disposed on the interior surface of said body between said valve seat and the outlet, and extending into the passageway; and a valve seat sealing portion disposed in fluid communication with said valve seat between the inlet and said flow limiter, and configured to seal said valve seat against the flow of fluid in the passageway when the fluid has a pressure that is less than a predetermined pressure.
2. The relief valve according to claim 1, wherein the predetermined pressure is less than about 1000 pounds per square inch.
3. The relief valve according to claim 2, wherein the predetermined pressure is less than about 500

pounds per square inch.

4. The relief valve according to claim 3, wherein the predetermined pressure is between about 200 and about 395 pounds per square inch.

5. The relief valve according to claim 1, wherein said flow limiter is configured to limit the flow of fluid through it to less than about 0.4 gallons per minute at a fluid pressure.

6. The relief valve according to claim 5, wherein said flow limiter is configured to limit the flow of fluid through it to from about 0.25 gallons per minute to about 1.25 gallons per minute.

7. The relief valve according to claim 5, wherein said flow limiter is comprised of a plurality of orifices.

8. The relief valve according to claim 1, wherein the valve seat is disposed generally parallel to the exterior surface of said body.

9. The relief valve according to claim 1, wherein the valve seat is disposed generally transverse to the exterior surface of said body.

10. The relief valve according to claim 1, wherein the valve seat sealing portion comprises a burst disk.

11. The relief valve according to claim 10, wherein the burst disk has a score.

12. A compact, pipe-burst-prevention valve system for use in a building plumbing system, comprising: a faucet housing having a handle and an outlet for delivering fluid to an end user; a control valve having an inlet and an outlet in fluid communication with each other and the control valve outlet in fluid communication with the faucet outlet; and a relief valve, comprising: a body having an inlet in fluid communication with the control valve inlet, an outlet in fluid communication with the control valve outlet, an exterior surface, and an interior surface that defines a generally longitudinal passageway that extends from the relief valve body inlet to the relief valve body outlet; a valve seat disposed on the interior surface of said body between the relief valve body inlet and outlet; a flow limiter disposed on the interior surface of said body between said valve seat and the relief valve body outlet and extending into the passageway; and a valve seat sealing portion disposed in fluid communication with said valve seat between the relief valve body inlet and said flow limiter, and configured to seal said valve seat against the flow of fluid when the fluid has a pressure that is less than a predetermined pressure.

13. The valve system according to claim 12, wherein said a control valve has a housing and said relief valve is disposed in the control valve housing.

14. The valve system according to claim 12, wherein the relief valve flow limiter is comprised of a plurality of orifices.

15. The valve system according to claim 12, wherein the valve seat is disposed generally parallel to the exterior surface of said relief valve body.

16. The relief valve according to claim 12, wherein the valve seat is disposed generally transverse to the exterior surface of said relief valve body.

17. The valve system according to claim 12, further comprising: a second control valve having an inlet and an outlet in fluid communication with each other and the second control valve outlet in fluid communication with the faucet outlet.

18. A method of preventing pipes from bursting, comprising the steps of: disposing a pressure relief valve in fluid communication with a building plumbing system, the pressure relief valve, comprising: a body having an inlet, an outlet, an exterior surface, and an interior surface that defines a generally longitudinal passageway that extends from the inlet to the outlet and is capable of allowing fluid to flow from inlet to outlet; a valve seat disposed on the interior surface of said body between the inlet and the outlet; a flow limiter disposed on the interior surface of said body between said valve seat and the outlet, and extending into the passageway; and a valve seat sealing portion disposed in fluid communication with said valve seat between the inlet and said flow limiter, and configured to seal said valve seat against the flow of fluid in the passageway when the fluid has a pressure that is less than a predetermined pressure.

19. The method according to claim 18, wherein the pressure relief valve is disposed in fluid communication with the building plumbing system between a fluid source and a fixture for delivering water to an end user.

20. The method according to claim 18, wherein the relief valve flow limiter is comprised of a plurality of orifices.
